



# GridShield

*Civil-Engineering-Aware Power Grid Advisor*

IBM Bob AI Hackathon 2026 · Challenge U1: Power Outage Prediction & Grid Equipment Failure Advisor

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**Team CivilGrid**

*Advisory only — human engineer sign-off required before any field action*

# The Problem

Grid operators must decide which assets to inspect or repair first — but the signals that matter live in disconnected silos.

- Electrical condition (temperature, vibration, oil quality) is tracked separately from civil-engineering condition (drainage, corrosion, foundation tilt).
- Weather forecasts are rarely fused with asset-level risk, so storm-driven failures catch crews off guard.
- Without a unified view, maintenance stays reactive and calendar-based instead of targeted and preventive.
- Money and crew time get spent on the wrong assets — while the truly critical ones wait.



**Electrical**



**Civil / Structural**



**Weather**

*Tracked in isolation today*

# Our Solution: GridShield

A transparent, rule-based advisory system that helps grid operators decide which assets to inspect or repair first — before a failure occurs.



## Electrical Sensor Telemetry

Temperature residual, vibration RMS, partial-discharge count, oil quality index, loading ratio



## Civil-Engineering Context

Drainage quality, erosion index, foundation tilt, corrosion score, vegetation clearance, asset age



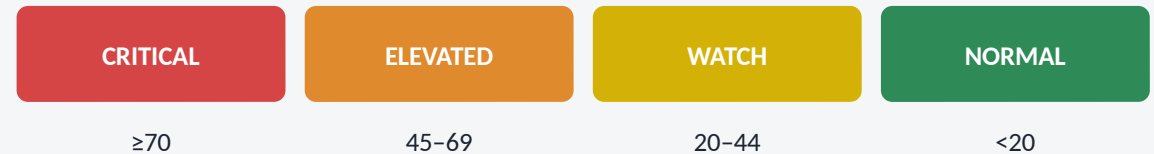
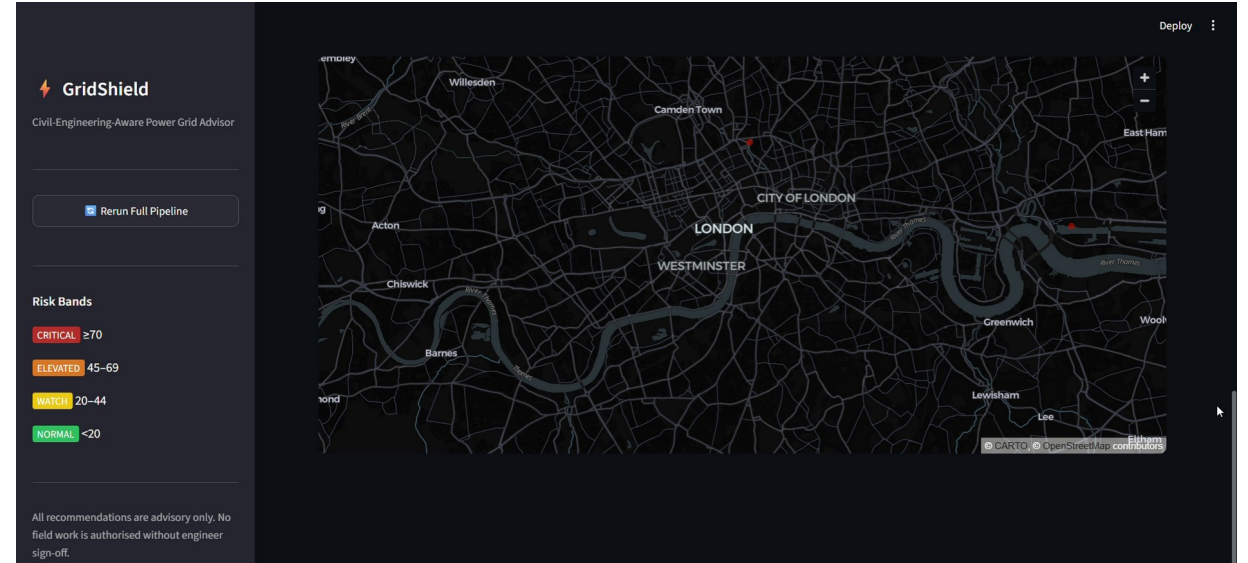
## 3-Day Weather Forecast

Wind gusts, rainfall, flood depth, lightning risk

13 auditable factors → **risk\_score (0-100)** → **priority\_score** (expected loss ÷ intervention cost)

# See It In Action: Ranked Risk Table

- Every asset is scored live under a selectable weather scenario (e.g. Storm Alpha — Severe) and time horizon: 24h, 72h, or 7-day.
- Assets are ranked by `priority_score`, not raw risk — a cheap fix on a hospital feeder can outrank an expensive fix on a redundant line.
- Color-coded risk bands make triage instant: CRITICAL ( $\geq 70$ ), ELEVATED (45–69), WATCH (20–44), NORMAL ( $< 20$ ).
- Advisory only — human engineer approval is always required before any field action.



*Assets ranked by `priority_score` —  $\text{net benefit} \div \text{intervention cost}$*

# Drilling Into a Single Asset

Asset A-02 · Transformer · Feeder F-101 · Critical load: water\_treatment

100.0 / 100

Risk Score

79.16

Priority Score

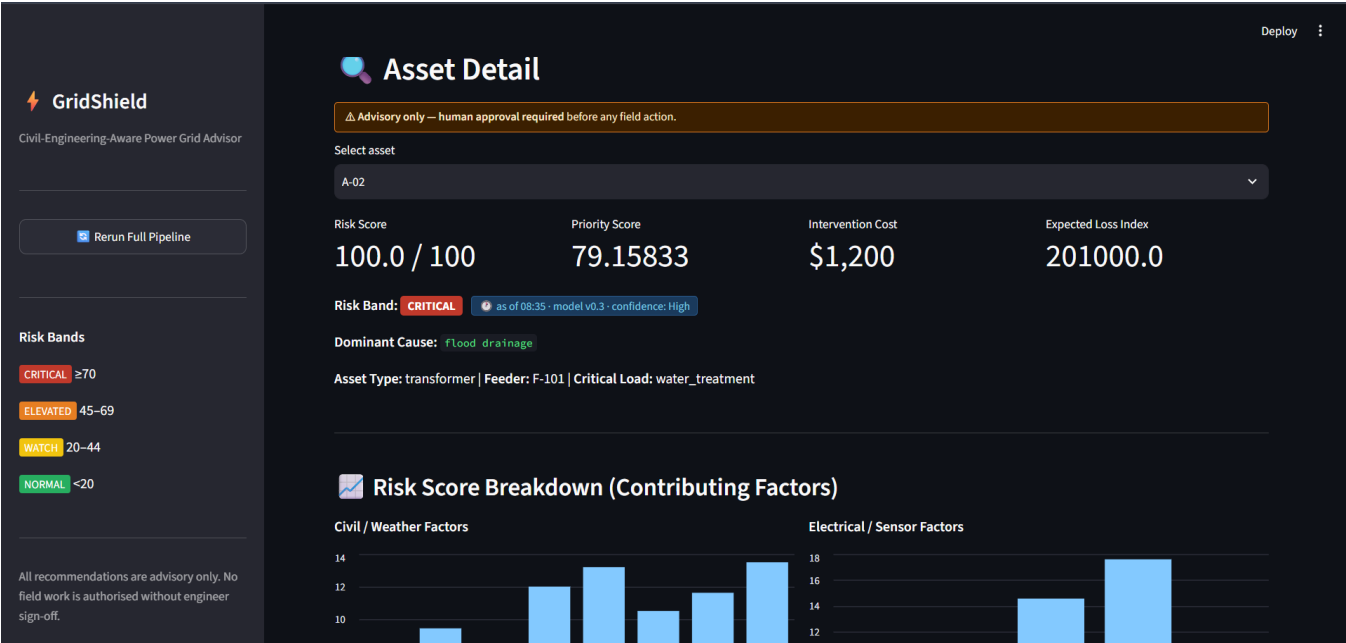
\$1,200

Intervention Cost

201,000

Expected Loss Index

Dominant cause: flood drainage



# How the Scoring Works

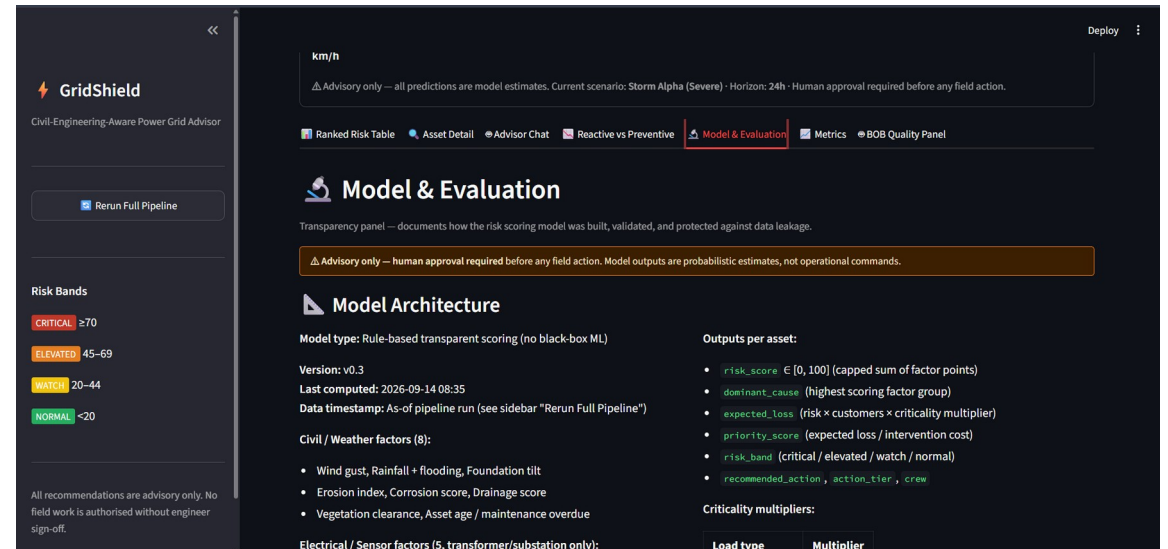
Rule-based, transparent scoring — no black-box ML • Model v0.3

## Civil / Weather factors (8)

- Wind gust
- Rainfall + flooding
- Foundation tilt
- Erosion index
- Corrosion score
- Drainage score
- Vegetation clearance
- Asset age / maintenance overdue

## Electrical / Sensor factors (5)

- Temperature residual
- Vibration RMS
- Partial-discharge count
- Oil quality index
- Loading ratio (transformer/substation only)



### Outputs per asset

risk\_score · dominant\_cause · expected\_loss · priority\_score · risk\_band · recommended\_action / action\_tier / crew

# Validated & Tested

**88.9%**

**Recall@Top-10**

8 of 9 seeded high-risk assets caught in the  
priority top-10

**38.25×**

**Loss Avoided per \$ Spent**

Every \$1 spent on preventive action avoids  
\$38.25 in expected loss

**PASS**

**Stress Test**

High-risk transformers stay in the top-10 even  
under a calm-weather baseline — risk isn't just  
weather-driven

*A ratio > 1.0 means the preventive queue generates more than \$1 of avoided loss per \$1 spent — the core economic argument for GridShield.*

# Built With IBM Bob, End to End

S-03

Added 6 electrical sensor fields, 5 new scoring factors, and 6 unit tests for transformer/substation risk.

S-04

Built the conversational advisor: comparison logic, a control-request refusal gate, a grounded timestamp footer, and 6 unit tests.

S-05

Polished the app (data-health badge, weather selector, crew card, metrics page) and wrote all docs — README, model card, safety case, evaluation report, data dictionary.

**Bob caught real defects during development** — e.g. a missing `newline=""` bug in `write_csv` causing double CRLF newlines on Windows (session S-01).

## Quality checklist — all PASS

- ✓ BOB panel exists in-app
- ✓ Crew pre-positioning shows location/ETA/materials
- ✓ Priority formula matches corrected version
- ✓ Metrics panel shows PR-AUC, Recall@Top-10, Brier, F1
- ✓ README, model card, safety case, evaluation report, data dictionary all present
- ✓ Live demo reset restarts pipeline from seed

# Known Limitations & What's Next

| Limitation             | Severity | Upgrade Path                                                    |
|------------------------|----------|-----------------------------------------------------------------|
| Synthetic sensor data  | Medium   | Data dictionary documents schema; real IoT adapter layer needed |
| Illustrative USD costs | Medium   | Calibrate intervention costs with real utility operations data  |
| 3-day forecast window  | Low      | Ingest hourly weather API for tighter storm timing              |
| Rule-based, not ML     | Medium   | Ensemble (gradient-boosted) model planned for v1.0              |

*All limitations are documented transparently in-app and in the model card. GridShield is advisory only — human engineer sign-off is always required before field action.*



# Thank You

*GridShield — Civil-Engineering-Aware Power Grid Advisor*

Team CivilGrid

[github.com/mitesh1502/bob-ai-hackathon-civilgrid](https://github.com/mitesh1502/bob-ai-hackathon-civilgrid)